

# STT 861: Theory of Probability & Statistics I

## Lecture 2: Set Theory II & Counting I

Guanqun Cao

Department of Statistics & Probability  
Michigan State University

Sep 03, 2026

Reference: Casella & Berger Ch. 1.1–1.2

# Lecture 2 Outline

## Today

- Recap of set operations
- Symmetric difference; three-set Venn diagrams
- Counting: Fundamental theorem, ordered sampling
- Permutations

**Reading:** CB §1.1–1.2

# Additional Set Operations

## Set Difference & Symmetric Difference

**Difference:**  $B - A = BA^c$   
(elements in  $B$  but not  $A$ )

**Symmetric Difference:**  
 $B \triangle A = (B - A) \cup (A - B)$   
 $= (A \cup B) - (A \cap B)$   
(elements in exactly one of  $A, B$ )

## Numeric Example

$A = \{1, 2, 3\}, B = \{2, 3, 4, 5\}$   
 $B - A = \{4, 5\}$   
 $A \triangle B = \{1, 4, 5\}$

## Three-set Venn diagram regions:

For events  $A, B, C$  there are  $2^3 = 8$  regions:

| Region          | Expression                  |
|-----------------|-----------------------------|
| Only $A$        | $ABC^c$ ... wait, $AB^cC^c$ |
| Only $B$        | $A^cBC^c$                   |
| Only $C$        | $A^cB^cC$                   |
| $A \cap B$ only | $ABC^c$                     |
| $A \cap C$ only | $AB^cC$                     |
| $B \cap C$ only | $A^cBC$                     |
| All three       | $ABC$                       |
| None            | $A^cB^cC^c$                 |

# Recap: Partition and Containment

## Key Relationships

- ①  $A \subset B$  iff  $A = AB$  iff  $B^c \subset A^c$
- ②  $A = B$  iff  $A \subset B$  and  $B \subset A$
- ③  $A = A \cup \emptyset = A \cap S$ ;  $\emptyset^c = S$ ;  $S^c = \emptyset$
- ④  $(A^c)^c = A$

## Proof of (1)

**Claim:**  $A \subset B \Leftrightarrow B^c \subset A^c$ .

( $\Rightarrow$ ): Let  $x \in B^c$ , so  $x \notin B$ . Since  $A \subset B$ , if  $x \in A$  then  $x \in B$  — contradiction. So  $x \notin A$ , hence  $x \in A^c$ .

( $\Leftarrow$ ): Suppose  $B^c \subset A^c$ . Let  $x \in A$ . Then  $x \notin A^c$ , so  $x \notin B^c$  (since  $B^c \subset A^c$ ), hence  $x \in B$ .  $\square$

# Why Counting?

## Classical Probability

When all outcomes are *equally likely*:

$$P(A) = \frac{\# \text{ outcomes in } A}{\# \text{ outcomes in } S} = \frac{|A|}{|S|}$$

Computing  $|A|$  and  $|S|$  requires **counting**.

## Motivating Problems

- How many ways to arrange 6 friends in a row for a photo?
- How many 4-digit PINs have no repeated digits?
- In a group of 10, how many committees of 3 are possible?
- What is  $P(\text{full house in 5-card poker})$ ?

# Classical (Equally Likely) Probability

## Setting

In many applications, probability describes the *relative frequency* at which events occur. The simplest setting:

- A **finite** sample space  $S$  with  $N$  points.
- All  $N$  sample points are **equally likely**.

Then  $P(\text{any single point}) = \frac{1}{N}$ .

## Classical Probability Formula

If  $A$  is an event and  $n(A)$  = number of sample points in  $A$ , then

$$P(A) = \frac{n(A)}{N} = \frac{\text{number of outcomes in } A}{\text{total number of outcomes}}$$

**All one has to do is count points in  $A$ .**

# Fundamental Theorem of Counting

## Theorem 1.2.14 (CB)

If a job consists of  $k$  separate tasks, and the  $i$ -th task can be done in  $n_i$  ways ( $i = 1, \dots, k$ ), then the entire job can be done in

$$n_1 \times n_2 \times \cdots \times n_k \text{ ways.}$$

## Applications

- **Experiment:** 2 temperatures  $\times$  3 pressures  $\times$  4 humidities = 24 combinations.
- **Passwords:** 8 characters, each from 62 possibilities (a-z,A-Z,0-9) =  $62^8 \approx 2.18 \times 10^{14}$ .
- **Flip 3 coins:**  $2^3 = 8$  outcomes.
- **Wardrobe:** 5 shirts  $\times$  3 pants  $\times$  2 shoes = 30 outfits.

# Ordered Sampling I: With Replacement

Consider a set  $\mathcal{A} = \{A_1, \dots, A_N\}$  of  $N$  distinct objects.

## Ordered Sample with Replacement

An **ordered sample of size  $r$  with replacement** from  $N$  objects:

Each draw is from the full set (same object can be chosen twice).

$$\text{Number of ordered samples} = N^r$$



## Examples

- PINs (4 digits, 0–9):  $10^4 = 10,000$
- Braille: 6 dots, each raised or flat:  $2^6 = 64$  patterns
- Toss coin  $n$  times:  $2^n$  outcomes

## Intuition

Each of the  $r$  draws independently comes from  $N$  choices. By the fundamental theorem:

$$N \times N \times \cdots \times N = N^r$$

# Ordered Sampling II: Without Replacement – Permutations

## Ordered Sample without Replacement

An **ordered sample of size  $r$  without replacement** from  $N$  distinct objects:

Once an object is drawn, it is not returned.

$$\text{Number of samples} = N(N-1)(N-2) \cdots (N-r+1) = \frac{N!}{(N-r)!} =: P(N, r)$$

## Factorial

$$N! = N \times (N-1) \times \cdots \times 2 \times 1 \quad \text{and} \quad 0! = 1.$$

$P(N, r)$  is called the number of **permutations** of  $r$  from  $N$ .

## Examples

- Arrange 6 friends in a row:  $P(6, 6) = 6! = 720$
- Seat 6 friends at a *round* table:  $(6 - 1)! = 5! = 120$  (fix one seat)
- Choose President, VP, Secretary from 10:  $P(10, 3) = 10 \times 9 \times 8 = 720$

# Sampling: Two Key Questions

## Two Key Choices in Counting

Many probability problems involve repeated selections from the same set. The number of possible patterns depends on two choices:

- ❶ **Replacement:** Are items put back after being selected?
  - *With replacement:* the same item can appear more than once.
  - *Without replacement:* each item can appear at most once.
- ❷ **Order:** Does the order of selection matter?
  - *Ordered:*  $(A_1, A_2) \neq (A_2, A_1)$ .
  - *Unordered:*  $\{A_1, A_2\} = \{A_2, A_1\}$ .

# Computing Probabilities with Permutations

## Bookshelf Problem (CB)

Mr. Jones has 10 books: 4 math (M), 3 chemistry (C), 2 history (H), 1 language (L). He wants all books of the same subject together. How many arrangements?

**Step 1:** Arrange the 4 subjects:  $4! = 24$  orderings of groups.

**Step 2:** Within each group:  $4! \times 3! \times 2! \times 1! = 24 \times 6 \times 2 \times 1 = 288$ .

**Total:**  $4! \times 4! \times 3! \times 2! \times 1! = 24 \times 288 = \mathbf{6,912}$ .

## Probability of a Specific Arrangement

If 10 books are placed randomly on a shelf, what is  $P(\text{all math books together})$ ?

**Favorable:** Treat 4 math books as 1 block  $\rightarrow$  7 objects,  $7!$  arrangements  $\times 4!$  internal arrangements.

**Total:**  $10!$

$$P = \frac{7! \times 4!}{10!} = \frac{5040 \times 24}{3628800} = \frac{120960}{3628800} \approx 0.0333$$

# Implication Notation and Sub-Events

## Sub-event / Implication

$A \subset B$  means “ $A$  implies  $B$ ”: knowing  $A$  occurred tells us  $B$  also occurred.

**Example:**  $A = \{\text{roll a 6}\}$ ,  $B = \{\text{roll an even}\}$ .

$A \subset B$ : rolling a 6 implies rolling an even. ✓

## Ordering arrangements with indistinguishable objects

Consider  $a$  objects of type A and  $b$  objects of type B (within each type, objects are indistinguishable).  
Number of distinct arrangements of all  $a + b$  objects:

$$\binom{a+b}{a} = \frac{(a+b)!}{a! b!}$$

**Example:** Arrange 4 M's and 3 I's in a row:  $\binom{7}{4} = 35$  ways.

# The 100 Prisoners Dilemma — The Problem

## Setup

- ➊ **100 prisoners** are each assigned a unique number 1–100.
- ➋ **100 boxes** are placed in a room, numbered 1–100. Each box contains one slip with a *random* prisoner number (one slip per box, all shuffled).
- ➌ Each prisoner enters the room *alone* and may open **at most 50 boxes**.
- ➍ Prisoners **cannot communicate** after the process begins. They may agree on a strategy beforehand.
- ➎ **Goal:** every prisoner finds their own number. If even one fails, all are executed.

Can they survive? Is there any hope?

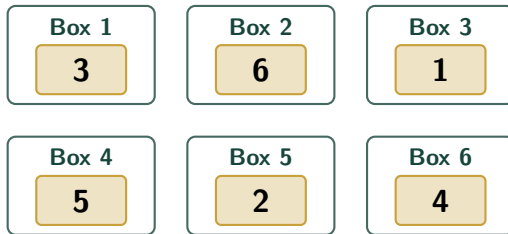
# The 100 Prisoners Dilemma — Illustration



**Prisoner  $i$**

opens  $\leq 50$  boxes

goal: find  
slip  $i$



*6 boxes shown (100 total)*

## Reading the diagram

Each box holds exactly one slip (a random permutation): Box 1  $\rightarrow$  slip 3; Box 2  $\rightarrow$  slip 6; Box 3  $\rightarrow$  slip 1; Box 4  $\rightarrow$  slip 5; ...



# The 100 Prisoners Dilemma — Naive Strategy Fails

## Strategy A: Open 50 Random Boxes

Each prisoner independently picks 50 boxes at random.

$$P(\text{prisoner } i \text{ finds their number}) = \frac{50}{100} = \frac{1}{2}$$

Since all 100 prisoners act **independently**:

$$P(\text{all 100 succeed}) = \left(\frac{1}{2}\right)^{100} \approx 8 \times 10^{-31} \approx \mathbf{0}$$

## Is there a better strategy?

- The prisoners **cannot** change the odds for any individual prisoner beyond 1/2 (they must open exactly 50 out of 100 boxes).
- But they *can* **correlate** their choices using a shared strategy agreed upon beforehand.

# The 100 Prisoners Dilemma — The Cycle Strategy

## Strategy B: Follow Your Cycle

Prisoner  $i$  proceeds as follows:

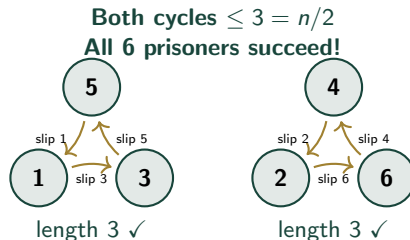
- 1 Open **Box**  $i$  (your own number).
- 2 Read slip  $j$  inside.
- 3 Open **Box**  $j$ .
- 4 Read the new slip, go to that box.
- 5 Repeat until you find **slip**  $i$  or you have opened 50 boxes.

## Why this is clever

Every permutation decomposes into **disjoint cycles**. Following the chain starting at box  $i$  always leads back to slip  $i$  — if the cycle is short enough.

**Worked example** (6 prisoners):

Permutation:  $1 \rightarrow 3 \rightarrow 5 \rightarrow 1$  and  $2 \rightarrow 6 \rightarrow 4 \rightarrow 2$



# The 100 Prisoners Dilemma — Probability Calculation

## Key Counting Result

The number of permutations of  $\{1, \dots, 100\}$  that contain a cycle of length exactly  $k$  is:

$$\binom{100}{k} (k-1)! \cdot (100-k)! = \frac{100!}{k}$$

So  $P(\text{there exists a cycle of length } k) = \frac{1}{k}$ .

## Computing the Success Probability

The strategy fails iff some cycle has length  $> 50$  (at most one such cycle can exist, since  $50+51 > 100$ ):

$$P(\text{failure}) = \sum_{k=51}^{100} \frac{1}{k} = \frac{1}{51} + \frac{1}{52} + \cdots + \frac{1}{100} \approx \ln 2 \approx 0.693$$

$$P(\text{all 100 succeed}) = 1 - \ln 2 \approx \mathbf{30.7\%}$$

| Strategy          | Individual $P$ | All succeed               |
|-------------------|----------------|---------------------------|
| Random (50 boxes) | $1/2$          | $\approx 10^{-30}$        |
| Follow the cycle  | $1/2$          | $\approx \mathbf{30.7\%}$ |

# In-Class Activity – Lecture 2

## Activity 2 – Set Theory & Counting

**Part A – Sets:** Let  $S = \{1, 2, \dots, 10\}$ ,  $A = \{1, 3, 5, 7, 9\}$ ,  $B = \{2, 3, 5, 7\}$ .

- (a) Find  $A \triangle B$  (symmetric difference).
- (b) Verify:  $|A \cup B| = |A| + |B| - |A \cap B|$ .
- (c) Do  $A$ ,  $A^c \cap B$ ,  $B^c \cap A^c$  partition  $S$ ? Prove or disprove.

## Part B – Counting:

- (d) A lock has 3 dials each with digits 0–9. How many combinations?
- (e) How many 3-letter words (A–Z) have no repeated letter?
- (f) A president, VP, and secretary are chosen from 8 candidates. How many ways?
- (g) **Bonus:** How many ways to arrange the letters in MISSISSIPPI?

## Activity 2 – Solutions

### Part A:

- (a)  $A \cap B = \{3, 5, 7\}$ ;  $A \triangle B = (A \cup B) - (A \cap B) = \{1, 2, 9\}$ ... actually  $A \cup B = \{1, 2, 3, 5, 7, 9\}$ , so  $A \triangle B = \{1, 2, 9\}$
- (b)  $|A| = 5, |B| = 4, |A \cap B| = 3; |A \cup B| = 6 = 5 + 4 - 3 \checkmark$
- (c)  $A \cup (A^c \cap B) \cup (B^c \cap A^c) = A \cup B \cup (A \cup B)^c = S \checkmark$ ; pairwise disjoint  $\checkmark$ . **Yes.**

### Part B:

- (d)  $10^3 = \mathbf{1000}$
- (e)  $P(26, 3) = 26 \times 25 \times 24 = \mathbf{15,600}$
- (f)  $P(8, 3) = 8 \times 7 \times 6 = \mathbf{336}$
- (g) MISSISSIPPI: 11 letters; M:1, I:4, S:4, P:2.  $\frac{11!}{1!4!4!2!} = \frac{39916800}{1152} = \mathbf{34,650}$